Replication of "Giving Mom a Break: The Impact of Higher EITC Payments on Maternal Health" by Evans and Garthwaite

Tate Mason
University of Georgia

March 19, 2026

Evans and Garthwaite's "Giving Mom a Break: The Impact of Higher EITC Payments on Maternal Health" is a paper which exploits variation in EITC expansions by family size. Their results find improvements in self-reported health for the women in their sample, as well as reductions in the likelihood of having risky biomarkers for the same women.

This report is focused on replicating their findings via their replication package submitted with the paper to AEJ. In addition to recreating the various figures and tables assigned, there is also an extension section at the end. The replication package is a product of its time. The code is poorly labeled, there are inconsistencies in variables used for regressions, and there is much room for improvement when considering best practices in organization of code.

My replications match except where noted in the assignment. These inconsistencies are likely due to sloppy reporting, using different model specifications than those listed, or simple rounding errors. I also attempt to better organize my files and code. The file tree has three primary branches: code, data, and output. The code folder contains each section's code and a main file to run each section. There are switches in the main file to allow the user to independently run sections by simply setting the switch to 0 if undesired or 1 if desired. There are also defined data and output paths in the main file, accessible across the section files. Turning to the data folder, there is a folder for all NHANES data and a single data file for the BRFSS data. Both are accessible by the aforementioned paths. Finally, output has subdirectories for graphs and tables to be saved into, as well as my log file in the core output directory.

1 Replication

Table 2: Sample Characteristics, Mothers Aged 21–40, 1993–1996 BRFSS

	$\leq$ High school education			College graduates		
Variable	1 child	2+ kids	p-value	1 child	2+ kids	p-value
Average age	31.0	32.046	0.000	32.249	34.622	0.000
% currently employed	0.7	0.580	0.000	0.797	0.714	0.000
Race						
% white, non-Hispanic	0.761	0.713	0.000	0.789	0.846	0.000
% black, non-Hispanic	0.129	0.141	0.014	0.104	0.069	0.000
% Hispanic	0.076	0.101	0.000	0.044	0.036	0.042
% other race	0.033	0.044	0.000	0.063	0.049	0.003
Marital status						
% married	0.548	0.655	0.000	0.709	0.846	0.000
% separated/divorced/widowed	0.238	0.210	0.000	0.162	0.120	0.000
% never married	0.184	0.111	0.000	0.111	0.025	0.000
Family income						
% $<20K$	0.405	0.386	0.006	0.110	0.082	0.000
% $\geq 20k, <50$K	0.422	0.441	0.007	0.440	0.411	0.003
% $\geq 50k$	0.078	0.086	0.038	0.374	0.432	0.000
% income missing	0.096	0.088	0.050	0.075	0.076	0.958
Health outcome						
% excellent/very good health	0.582	0.577	0.447	0.805	0.809	0.580
% with any bad mental health days	0.432	0.447	0.029	0.418	0.424	0.577
% with any bad physical health days	0.351	0.343	0.253	0.357	0.356	0.969
Number of bad mental health days	4.270	4.523	0.028	2.932	2.892	0.744
Number of bad physical health days	2.809	2.647	0.072	1.889	1.949	0.549
Observations	7,315	15,737		3,881	6,740	

Notes: The p-value is for the test of the null hypothesis that the means across the samples are the same. The test isperformed allowing for an arbitrary correlation for observations within a state.

Table 3: Difference-in-Differences OLS and Negative Binomial Estimates
Mothers Aged 21–40, 1993–2001 BRFSS

	Preexpansion mean (treat.)	Estimation method	DiD estimates	
			Simple	Reg. adjusted
At work	0.580	OLS	0.017	0.020
			(0.007)	(0.007)
			[0.023]	[0.008]
Excellent/very good health?	0.577	OLS	0.009	0.014
			(0.008)	(0.008)
			[0.233]	[0.078]
Number of bad mental health days in past month	4.523	Negative binomial	-0.047	-0.075
			(0.031)	(0.033)
			[0.121]	[0.021]
Number of bad physical health days in past month	2.647	Negative binomial	0.014	0.010
			(0.039)	(0.039)
			[0.719]	[0.788]

Notes: Standard errors are reported in parentheses and p-values on the test of the null that the coefficient is zeroare reported in square brackets. All standard errors allow for arbitrary correlations between observations within thesame state. Other covariates in the difference-in-differences model include: Complete set of dummies for age, race,marital status, and number of children for the respondent, plus a complete set of dummies for the month of survey,year of survey, and state of residence. All standard errors allow for arbitrary correlations within state. Regression-adjusted estimates include dummies for age, race, marital status, number of children, month, year, and state of residence.

Table 4: Robustness Tests, Women Aged 21–40, 1993–2001 BRFSS

Outcome	DD Method	DD state×year FE	2 vs 0 kids	DD married	DD single	DDD
At work (OLS)	0.0203	0.0206	0.0636	0.0183	0.0457	0.0130
	(0.0074)	(0.0076)	(0.0097)	(0.0109)	(0.0110)	(0.0123)
	[0.0085]	[0.0092]	[0.0000]	[0.0986]	[0.0001]	[0.2982]
Exc./very good health (OLS)	0.0136	0.0119	0.0148	0.0211	0.0082	-0.0045
	(0.0074)	(0.0075)	(0.0100)	(0.0088)	(0.0127)	(0.0114)
	[0.0730]	[0.1182]	[0.1457]	[0.0205]	[0.5212]	[0.6954]
Bad mental health days (NegBin)	-0.0687	-0.0615	-0.0195	-0.1027	-0.0514	-0.0622
	(0.0332)	(0.0326)	(0.0423)	(0.0519)	(0.0519)	(0.0730)
	[0.0385]	[0.0594]	[0.6444]	[0.0478]	[0.3221]	[0.3936]
Bad physical health days (NegBin)	0.0098	0.0249	-0.0499	0.0432	-0.0377	0.1226
	(0.0398)	(0.0391)	(0.0521)	(0.0508)	(0.0675)	(0.0911)
	[0.8055]	[0.5250]	[0.3384]	[0.3943]	[0.5766]	[0.1781]

Notes: Standard errors in parentheses; *p*-values in square brackets. All standard errors allow for arbitrary correlations within state.

Table 5: Biomarkers for Mothers Aged 21–40 with a High School Education or Less, NHANES

Biomarker	Obs	Mean	Risky level	% Risky
Measures of inflammation				
C-reactive protein (mg/Dl)	2,950	0.573	$\geq$ 0.3 mg/Dl	0.437
Albumin (g/Dl)	2,935	4.066	$<$ 3.8 g/Dl	0.262
Number of risky inflammation conditions	2,934	0.699		
Any risky inflammation condition	2,934			0.526
Measures of cardiovascular conditions				
Diastolic blood pressure (mmHg)	2,947	69.303	$\geq$ 140 mmHg	0.046
Systolic blood pressure (mmHg)	2,952	112.205	$\geq$ 90 mmHg	0.035
Resting pulse (beats/min)	3,090	74.975	$\geq$ 90 BPM	0.108
Number of risky cardiovascular conditions	2,947	0.185		
Any risky cardiovascular condition	2,947			0.155
Measures of metabolic conditions				
Total cholesterol (mg/Dl)	2,949	189.949	$\geq$ 240 mg/Dl	0.102
HDL (mg/Dl)	2,942	53.625	$<$ 40 mg/Dl	0.156
Glycated hemoglobin (%)	2,992	5.204	$\geq$ 6.4%	0.026
Number of risky metabolic conditions	2,933	0.284		
Any risky metabolic condition	2,933			0.259
Aggregate risks				
Number of risky conditions	2,683	1.156		
One or more risky conditions	2,683			0.657
Two or more risky conditions	2,683			0.333
Three or more risky conditions	2,683			0.127

Notes: Sample restricted to mothers aged 21–40 with a high school degree or less. NHANES III, 1999/2000, 2001/2002, 2003/2004.

Table 6: Regression-Adjusted DD and DDD Estimates for Effect of EITC Expansion on Allostatic Load, Women Aged 21–40

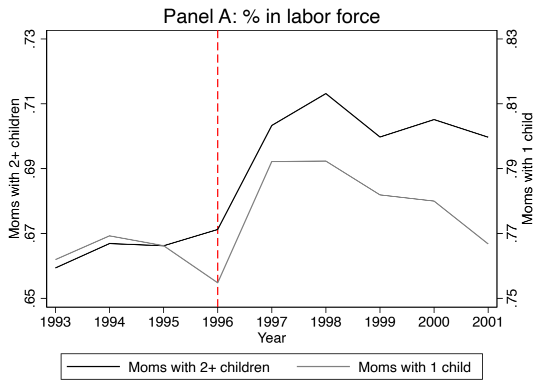
Outcome	Preexpansion mean (treatment group)	DD	DDD	Two Children vs. No Children
One or more risky conditions	0.640	-0.091	-0.093	-0.1481
		(0.050)	(0.072)	(0.077)
		[0.066]	[0.199]	[0.053]
Two or more risky conditions	0.311	-0.094	-0.054	-0.0310
		(0.053)	(0.071)	(0.084)
		[0.074]	[0.448]	[0.713]
Three or more risky conditions	0.113	-0.061	-0.035	-0.0334
		(0.039)	(0.053)	(0.070)
		[0.118]	[0.507]	[0.633]
Poisson: total risky conditions	1.110	-0.235	-0.207	-0.153
		(0.095)	(0.150)	(0.152)
		[0.014]	[0.169]	[0.314]

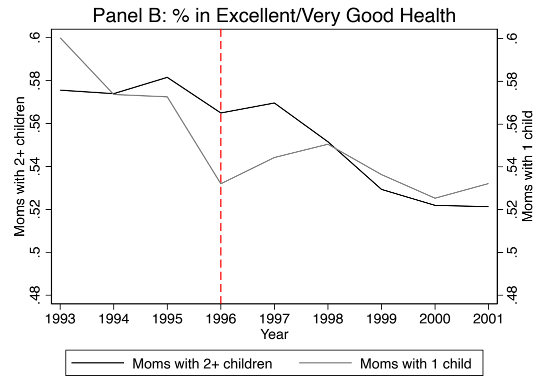
Notes: Standard errors in parentheses; p-values in square brackets. All standard errors allow for arbitrary heteroskedasticity.

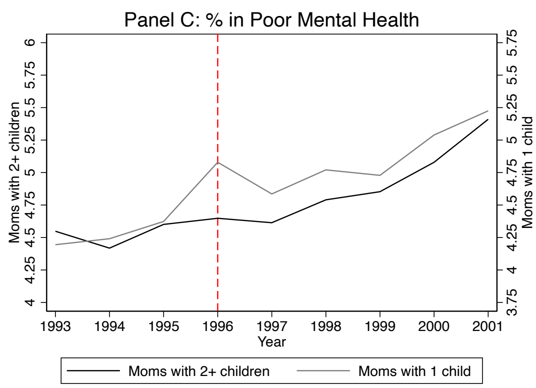
Table 7: Regression-Adjusted DD and DDD Estimates for Individual Biomarkers, Women Aged 21–40

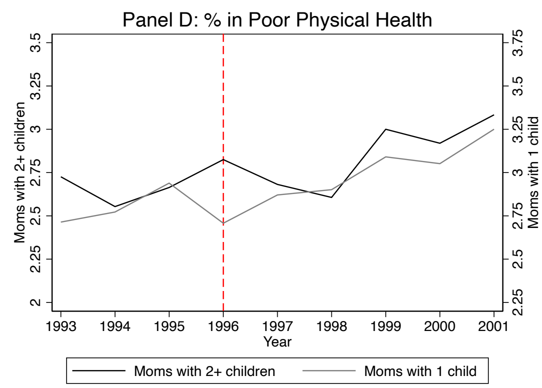
Outcome	Preexpansion mean (treatment group)	DD	DDD	Two Children vs. No Children
Panel A. Metabolic biomarkers				
Risky glycated hemoglobin	0.030	-0.004	-0.012	-0.048
		(0.013)	(0.019)	(0.032)
		[0.773]	[0.529]	[0.136]
Risky total cholesterol	0.082	-0.022	0.043	0.094
		(0.034)	(0.046)	(0.048)
		[0.520]	[0.351]	[0.051]
Risky HDL	0.172	-0.027	-0.044	-0.046
		(0.036)	(0.047)	(0.060)
		[0.453]	[0.348]	[0.440]
Any risky metabolic condition	0.256	-0.042	-0.017	0.044
		(0.045)	(0.061)	(0.072)
		[0.348]	[0.776]	[0.541]
Poisson: number of risky metabolic conditions		-0.185	0.030	0.005
		(0.177)	(0.276)	(0.290)
		[0.294]	[0.914]	[0.985]
Panel B. Cardiovascular biomarkers				
Risky diastolic blood pressure	0.062	-0.032	-0.030	0.015
		(0.017)	(0.026)	(0.035)
		[0.059]	[0.251]	[0.664]
Risky systolic blood pressure	0.040	0.004	-0.000	0.013
		(0.014)	(0.023)	(0.027)
		[0.800]	[0.984]	[0.620]
Risky pulse	0.068	-0.016	-0.043	-0.012
		(0.037)	(0.049)	(0.062)
		[0.669]	[0.381]	[0.842]
Any risky cardiovascular condition	0.134	-0.034	-0.048	0.003
		(0.041)	(0.055)	(0.072)
		[0.407]	[0.392]	[0.962]
Poisson: number of risky cardiovascular conditions		-0.317	-0.423	0.182
		(0.233)	(0.343)	(0.344)
		[0.174]	[0.217]	[0.596]
Panel C. Inflammation biomarkers				
Risky albumin	0.278	-0.088	-0.087	-0.124
		(0.045)	(0.063)	(0.073)
		[0.052]	[0.167]	[0.091]
Risky C-reactive protein	0.383	-0.083	-0.012	-0.040
		(0.050)	(0.070)	(0.079)
		[0.098]	[0.859]	[0.616]
Any risky inflammatory condition	0.493	-0.096	-0.060	-0.086
		(0.050)	(0.071)	(0.078)
		[0.057]	[0.402]	[0.271]
Poisson: number of risky inflammatory conditions		-0.217	-0.136	-0.216
		(0.099)	(0.159)	(0.158)
		[0.029]	[0.394]	[0.171]

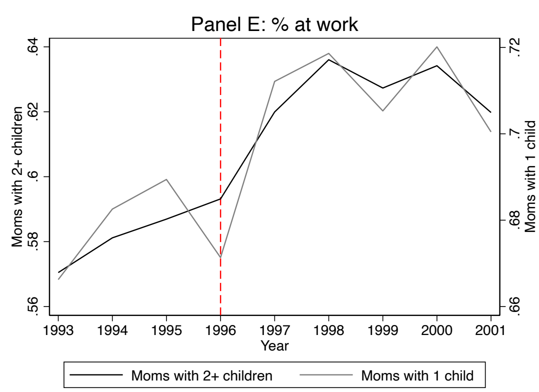
Notes: Standard errors in parentheses; p-values in square brackets. All standard errors allow for arbitrary heteroskedasticity. DD covariates: dummies for age, race, marital status, and survey year. DDD covariates add interactions between education×year, kids×year, and education×kids. Column 4 uses stacked NHANES data and compares women with 2+ children to women with no children.


Panel A: % in labor force
Moms with 2+ children
Moms with 1 child
Year
Moms with 2+ children
Moms with 1 child


Panel B: % in Excellent/Very Good Health
Moms with 2+ children
Moms with 1 child
Year
Moms with 2+ children
Moms with 1 child


Panel C: % in Poor Mental Health
Moms with 2+ children
Moms with 1 child
Year
Moms with 2+ children
Moms with 1 child


Panel D: % in Poor Physical Health
Moms with 2+ children
Moms with 1 child
Year
Moms with 2+ children
Moms with 1 child


Panel E: % at work
Moms with 2+ children
Moms with 1 child
Year
Moms with 2+ children
Moms with 1 child

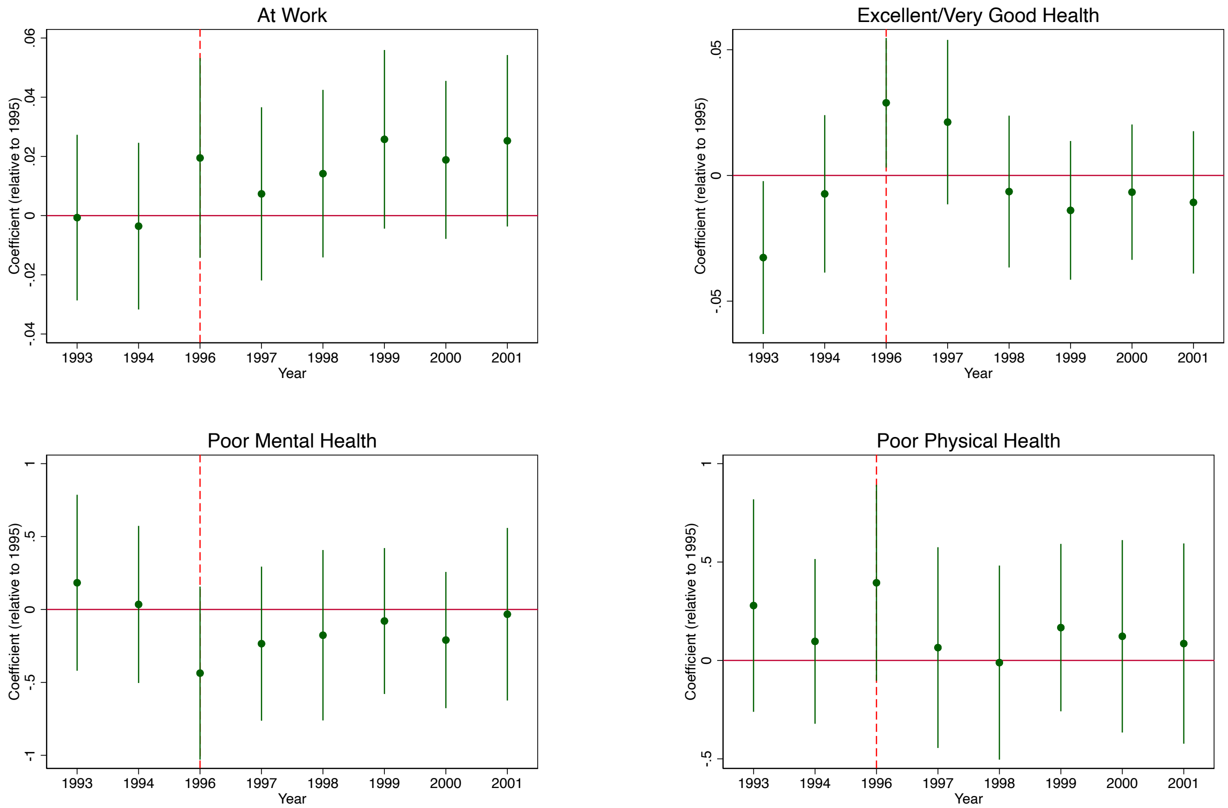
2 Additional Robustness Checks - Footnotes 12 and 21

Additional Robustness Checks, Women Aged 21–40, 1993–2001 BRFSS

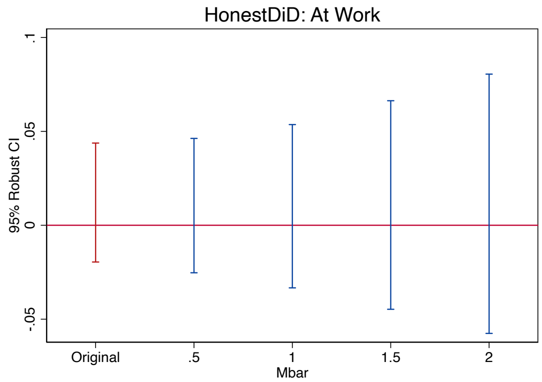
Outcome	1995 treatment (Footnote 12)	Excl. 2000+ (Footnote 21)	Excl. 1999+ (Footnote 21)
At work (OLS)	-0.015	0.034	0.034
	(0.008)	(0.007)	(0.009)
	[0.069]	[0.000]	[0.000]
Exc./very good health (OLS)	0.005	0.014	0.021
	(0.008)	(0.008)	(0.009)
	[0.540]	[0.063]	[0.017]
Bad mental health days (NegBin)	-0.000	-0.042	-0.054
	(0.040)	(0.030)	(0.033)
	[0.998]	[0.170]	[0.102]
Bad physical health days (NegBin)	0.023	-0.010	-0.035
	(0.063)	(0.037)	(0.041)
	[0.711]	[0.793]	[0.399]

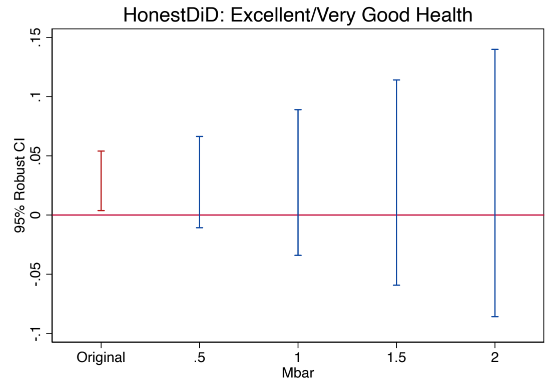
Notes: Standard errors in parentheses; p-values in square brackets. All standard errors allow for arbitrary correlations within state. Column 1 tests whether treatment in 1995 (rebates from tax year 1994) drives results. Columns 2–3 exclude years where the Child Tax Credit was in existence.

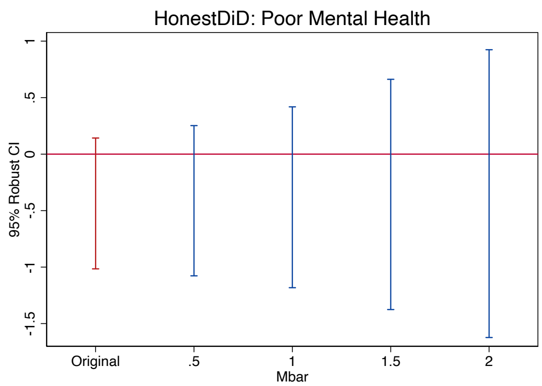
3 Extension - Event Study of Outcomes (1995 Normalized to Zero)

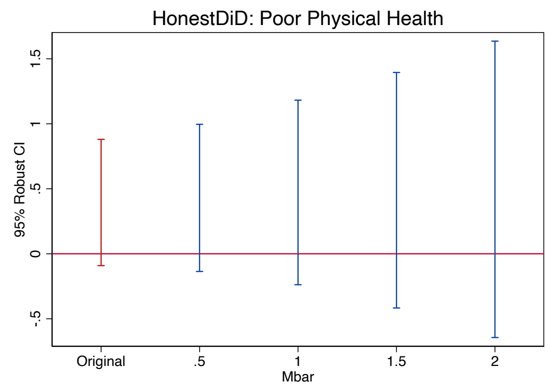


3.1 Parallel Trends Check









The event study rules out parallel trends for three of the outcomes. However, for the excellent/very good health reporting, it would appear that leading up to the expansion, self-reporting of good-to-excellent health was already increasing. There is a downward trend for presenting poor mental health as well, but the confidence interval encompasses zero, so it is a less compelling result. For working and poor physical health, there is evidence in favor of parallel trends. Both of these variables, relative to 1995, have little pre-event variation, validating the assumption of parallel trends.

When using the HonestDID package, we can see that there is indeed pre-trends in the excellent/very good health outcome. However, we can also rule out pre-trends for the other outcomes of interest, as they are all robust to pretrends to a threshold of $\bar{M} = 2$. When discussing the excellent/very good health pre-trends, it could potentially be due to the 1993 EITC expansion leading to already improving health for women with two children relative to those with only one child.